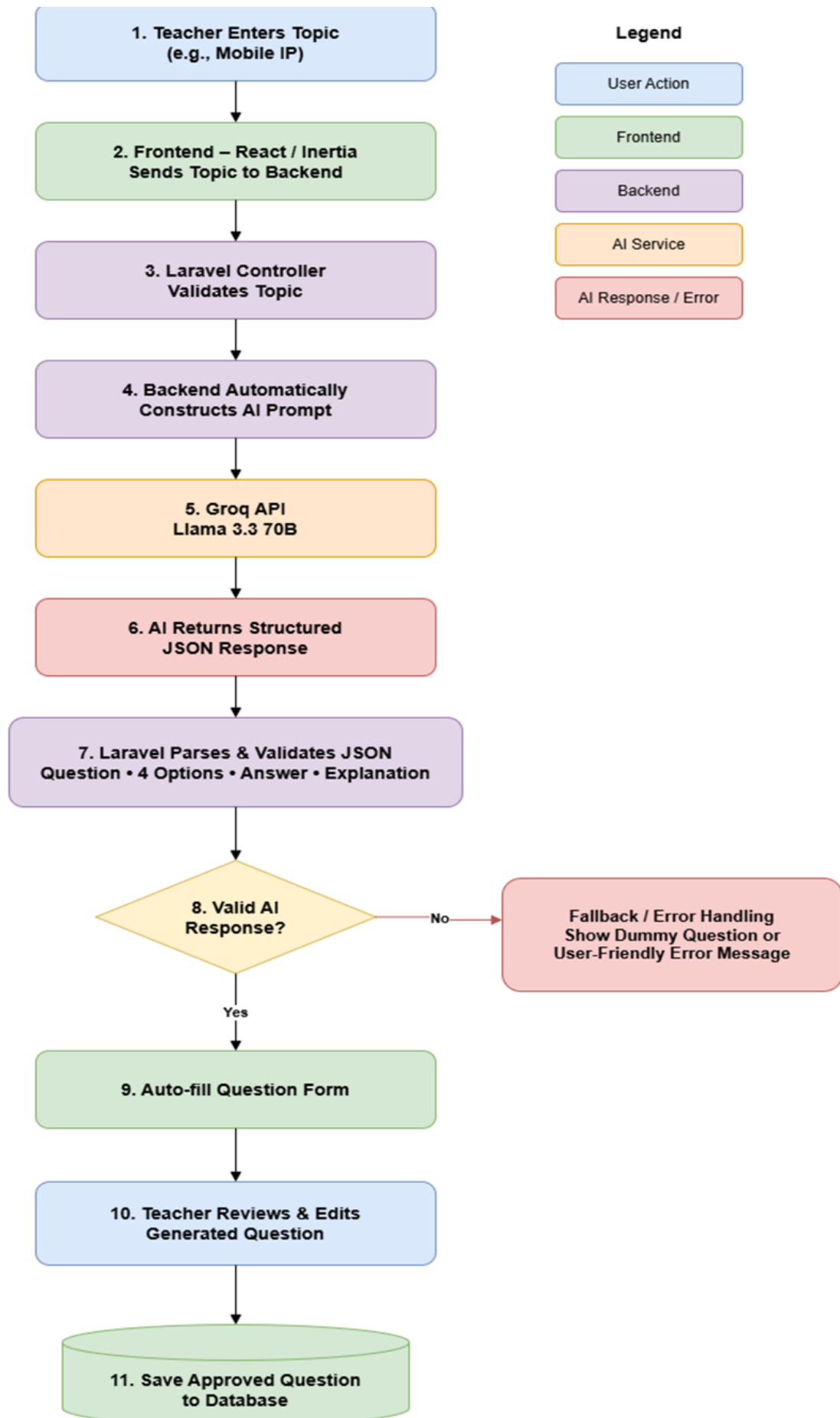


AI Workflow - AI-Based Smart Quiz System



Step-by-step Description

1. The teacher enters a topic, such as Mobile IP, in the AI question-generation interface. This topic serves as the main input for generating a relevant multiple-choice question.
2. The React/Inertia frontend captures the teacher's topic and sends it to the Laravel backend for further processing.
3. The Laravel controller receives the request and validates the topic to ensure that the input is present and suitable for processing.
4. After validation, the backend automatically constructs a detailed AI prompt using predefined instructions together with the teacher-provided topic.
5. The constructed prompt is sent to the Groq API, where the Llama 3.3 70B model processes the request and generates the required educational content.
6. The AI returns a structured JSON response containing the generated question, four answer options, the correct answer, and an explanation.
7. Laravel parses and validates the returned JSON. It checks the required fields, confirms that exactly four options are present, and verifies that the correct answer matches one of the options.
8. The system checks whether the AI response is valid. If the response is invalid, empty, malformed, or the AI service is unavailable, the system moves to fallback or error handling and displays an appropriate user-friendly message.
9. If the AI response is valid, the frontend receives the generated data and automatically fills the question form with the question, four options, correct answer, and explanation.
10. The teacher reviews the generated question and can edit the content when necessary before approving it.
11. After review and approval, the teacher saves the question, and the finalized question is stored in the database for use in the quiz system.